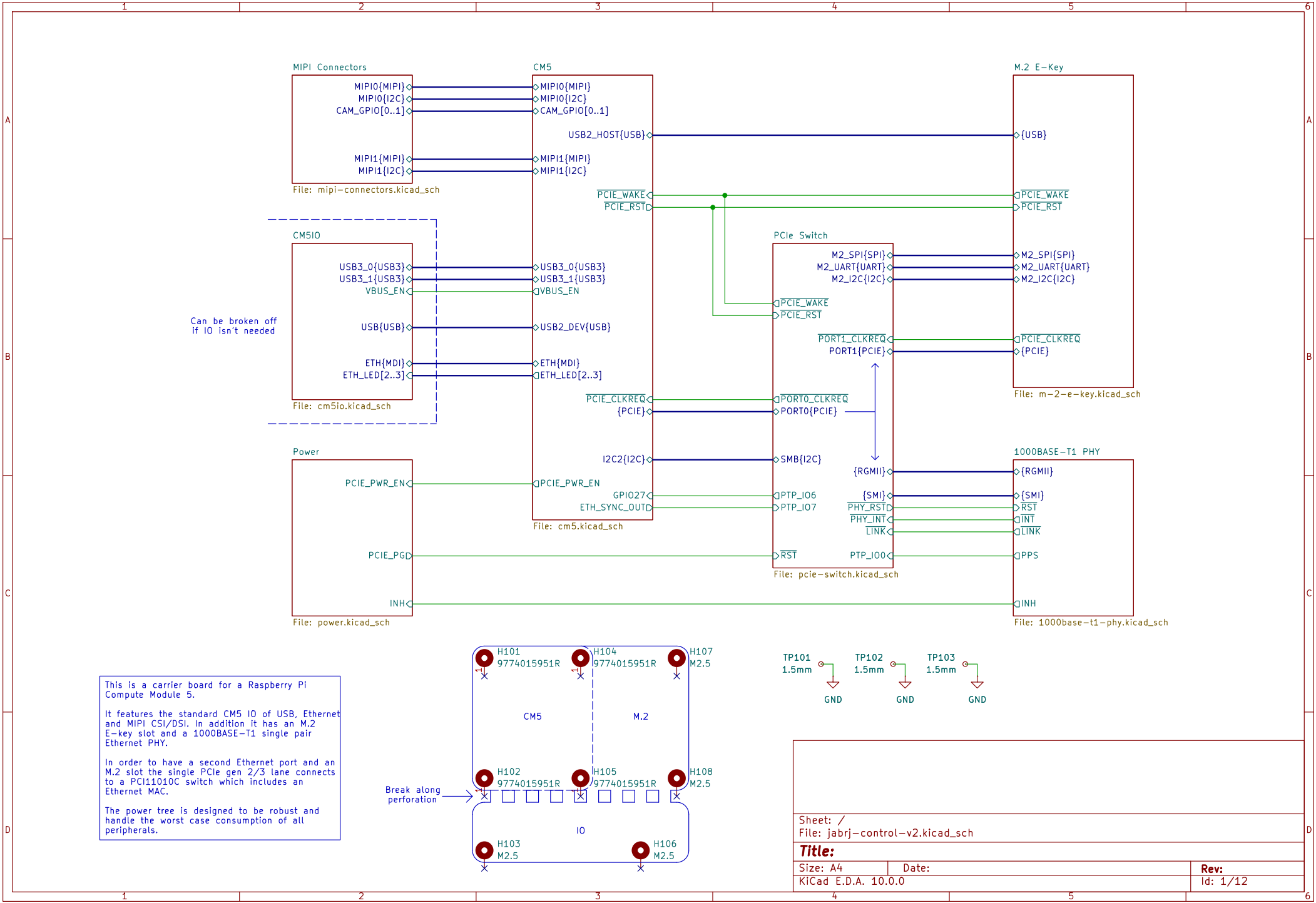




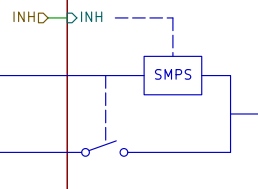
Power In



VIN (12V-24V)

<1750mA

5V Regulator



5V

CM5IO



VBUS

<4900mA

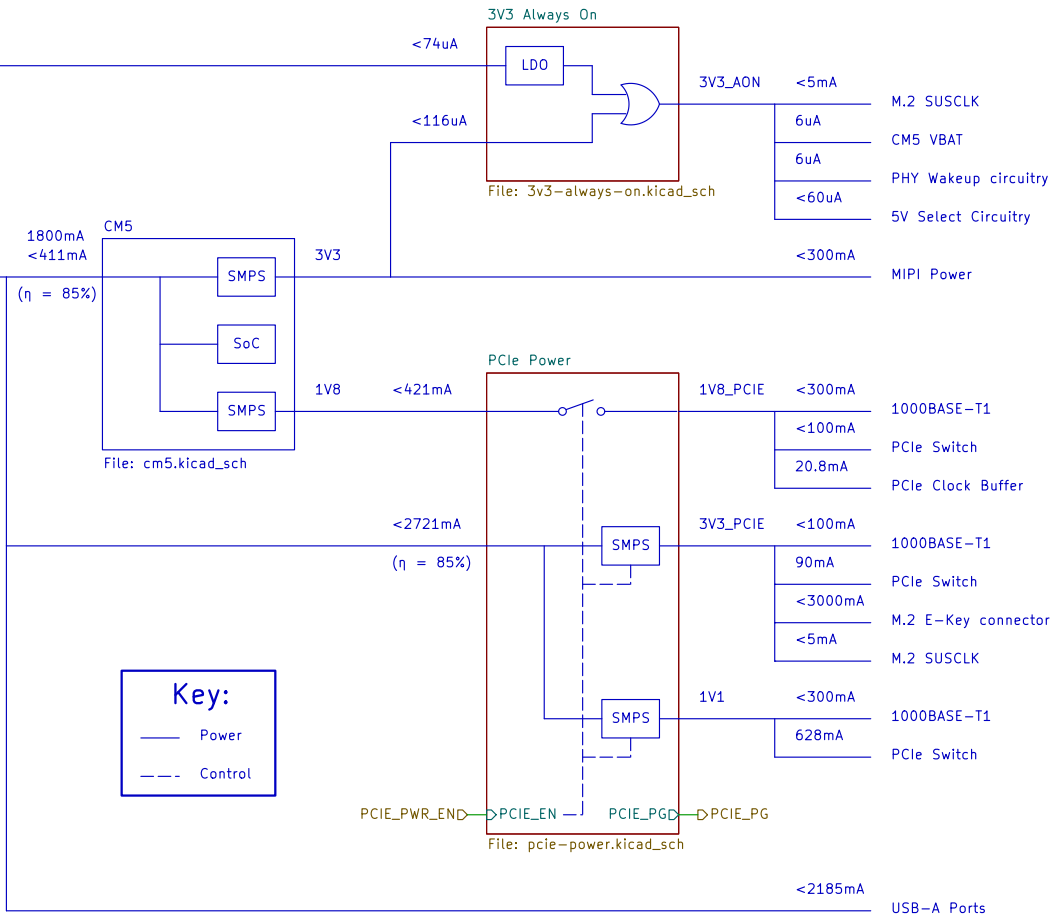
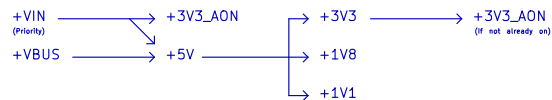
File: 5v-regulator.kicad_sch

The maximum power that can be consumed by the system is 13W, and the maximum power that could be consumed by peripherals (M.2 & USB-A ports) is 20.4W. Not all sources are able to provide this power so the peripheral power must be limited:

- 5V USB (3A capable): 2W for peripherals
- 5V USB (5A capable): 12W for peripherals
- 12V VIN: 5W for peripherals
- 24V VIN: 20W for peripherals

Note that the USB-A ports cannot consume more than 10W combined.

Power up sequence:



Key:

- Power
- Control

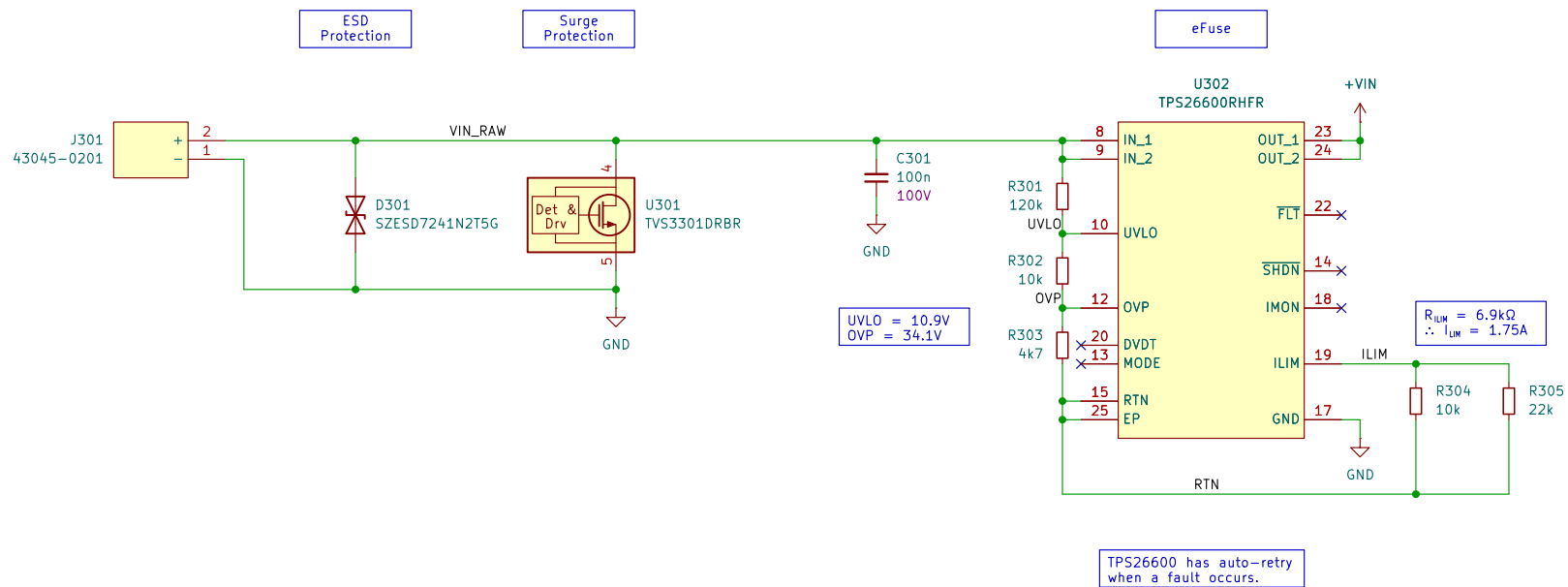
Sheet: /Power/
File: power.kicad_sch

Title:

Size: A4
KiCad E.D.A. 10.0.0

Date:

Rev:
Id: 2/12



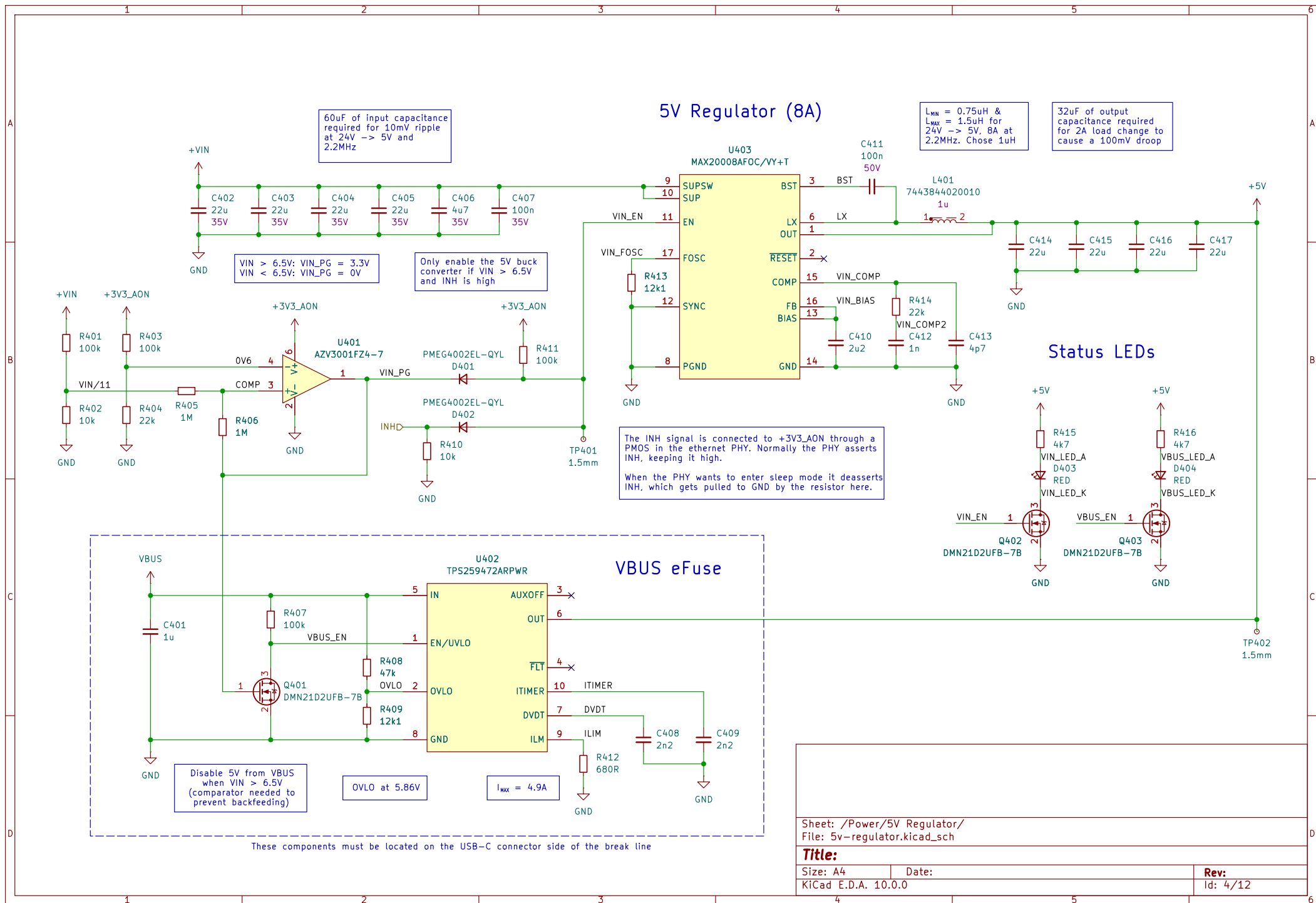
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File: power-in.kicad_sch

Title:

Size: A4 Date:

KiCad E.D.A. 10.0.0

Rev:
Id: 3/12



Sheet: /Power/5V Regulator/
File: 5v-regulator.kicad_sch

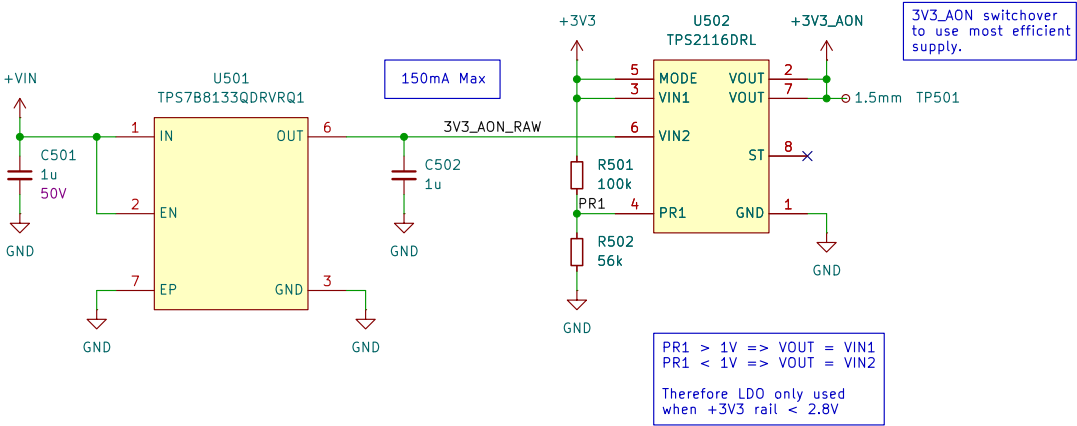
Title:

Size: A4
KiCad E.D.A. 10.0.0

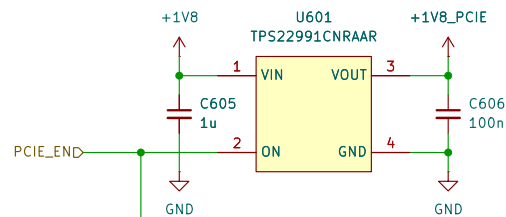
Date:

Rev:
Id: 4/12

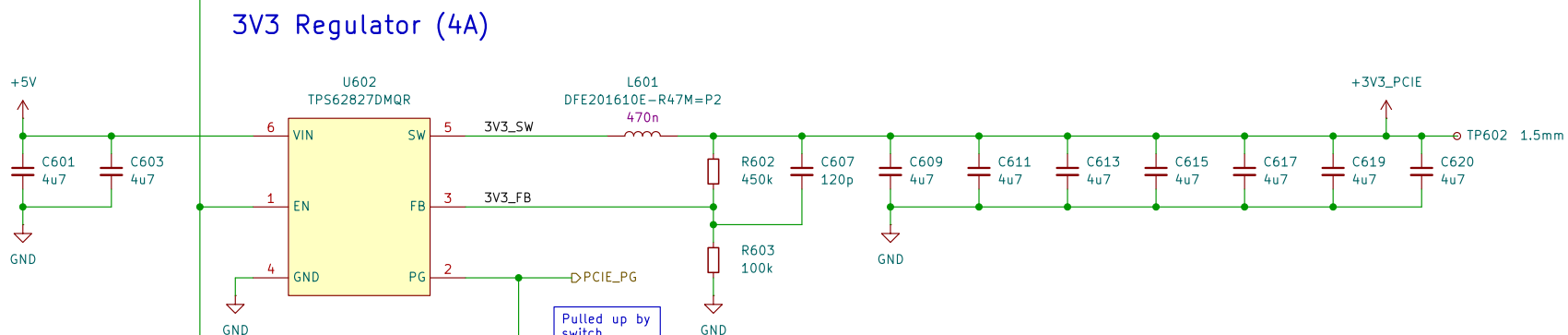
3V3 Always on supply



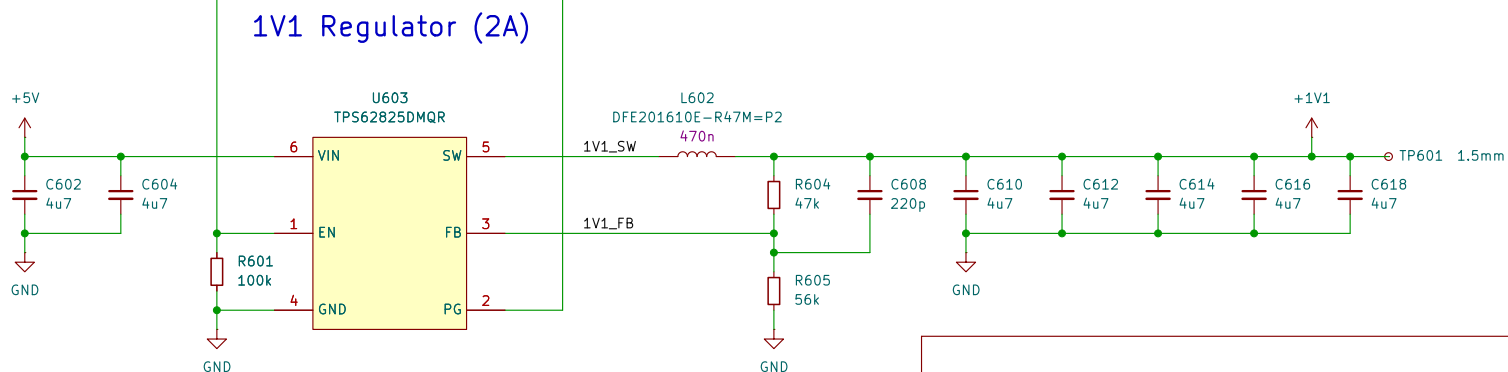
Sheet: /Power/3V3 Always On/ File: 3v3-always-on.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 5/12



1V8 Load Switch (3A)



3V3 Regulator (4A)



1V1 Regulator (2A)

Use high quality, low ESR (<50mΩ) capacitors such as GRM155R61A475ME02

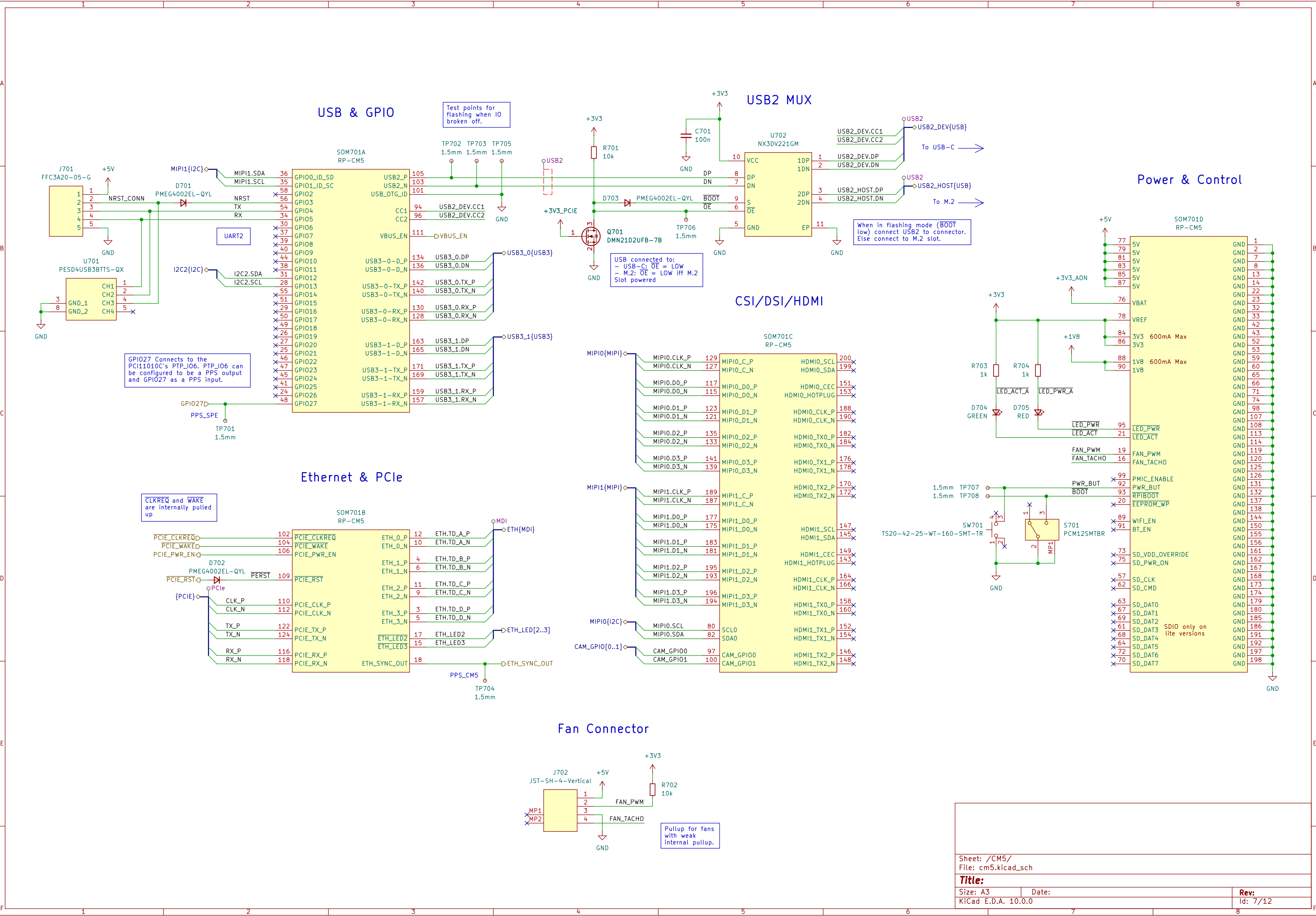
Sheet: /Power/PCIE Power/
File: pcie-power.kicad_sch

Title:

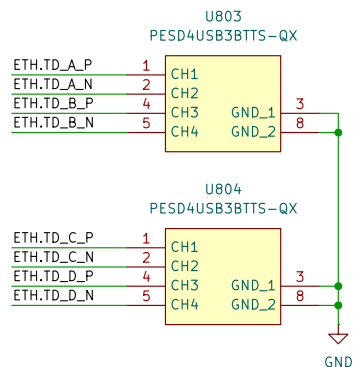
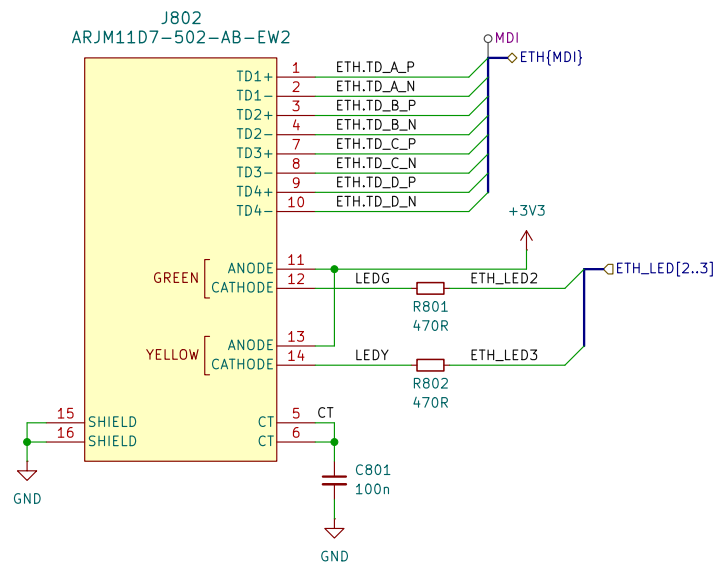
Size: A4 Date:

KiCad E.D.A. 10.0.0

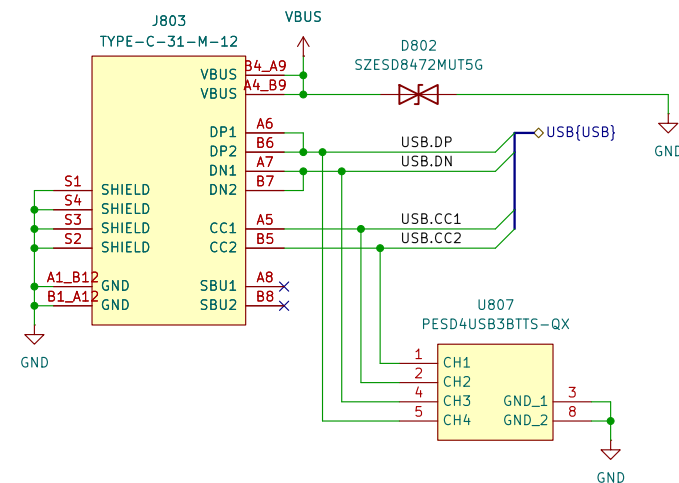
Rev:
Id: 6/12



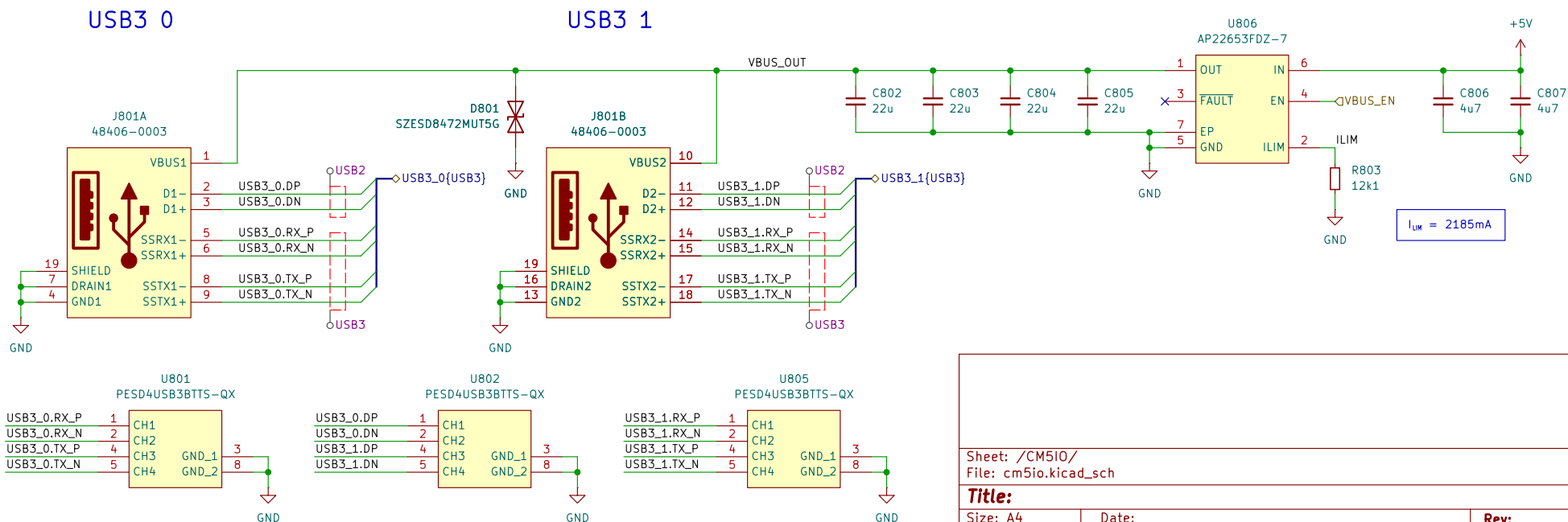
Ethernet RJ45



USB2



VBUS Load Switch



Sheet: /CM510/
File: cm510.kicad_sch

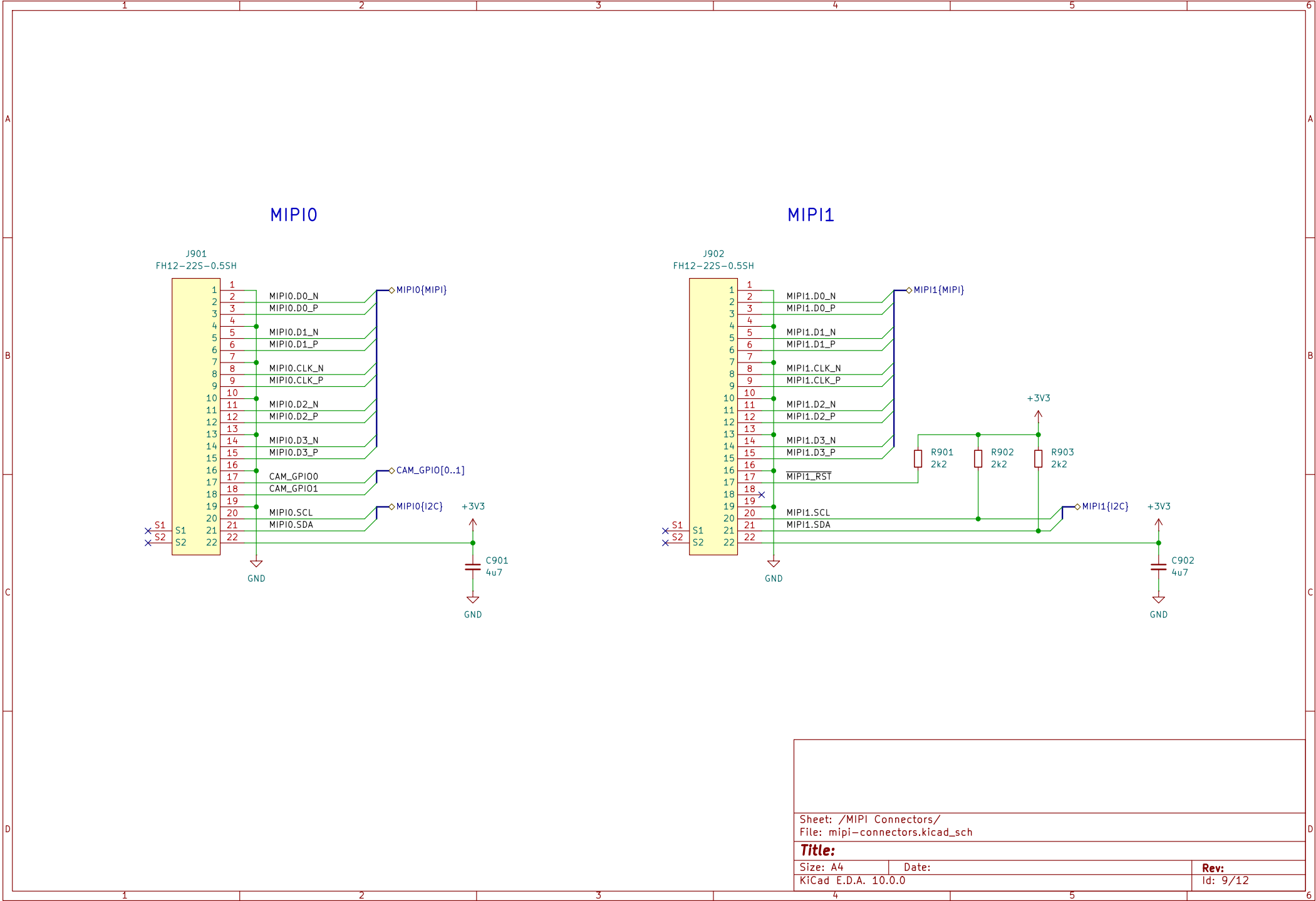
Title:

Size: A4 Date:

KiCad E.D.A. 10.0.0

Rev:

Id: 8/12



Sheet: /MIPI Connectors/
File: mipi-connectors.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.0

Rev:

Id: 9/12

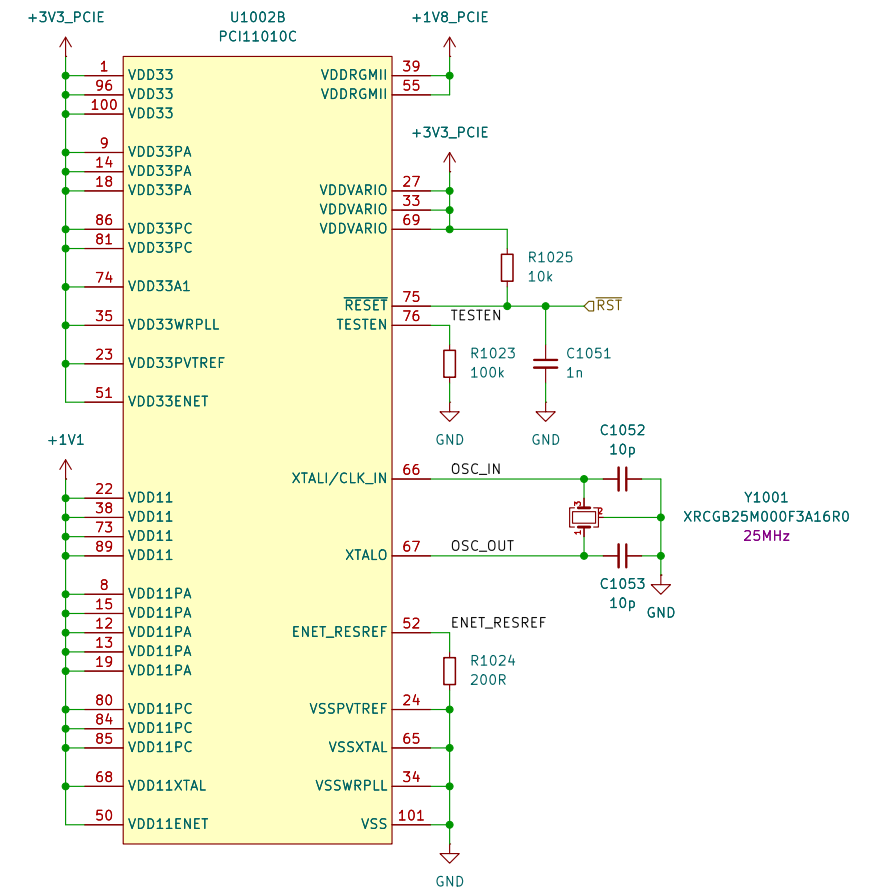
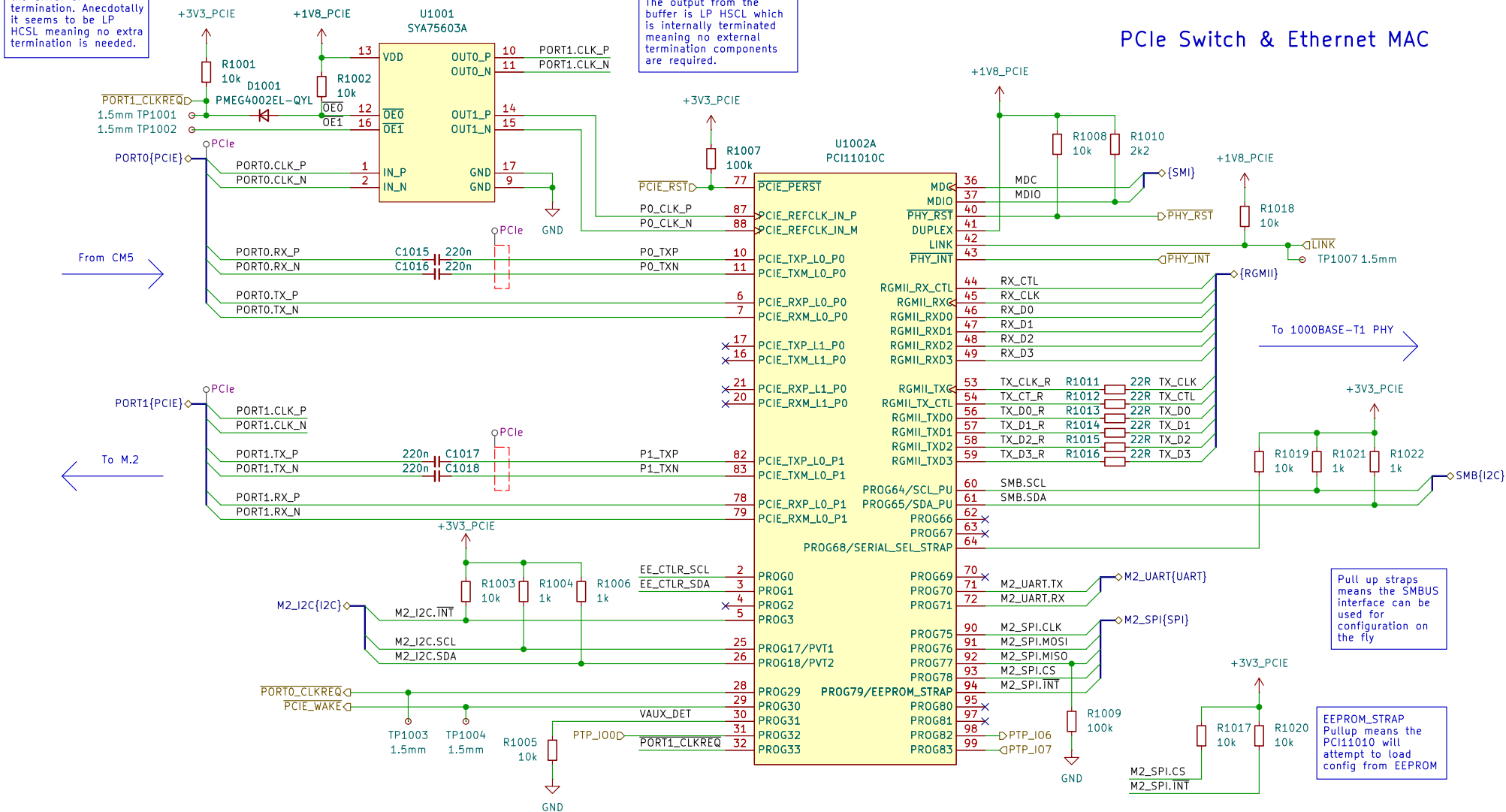
PCIe CLK termination should be done by the host, however Raspberry Pi / Broadcom doesn't give any information about the driver or termination. Anecdotaly it seems to be LP HCSL meaning no extra termination is needed.

OE pins have internal 100k pull down resistors. Diode used to level shift from 3.3V to 1.8V (to save power)

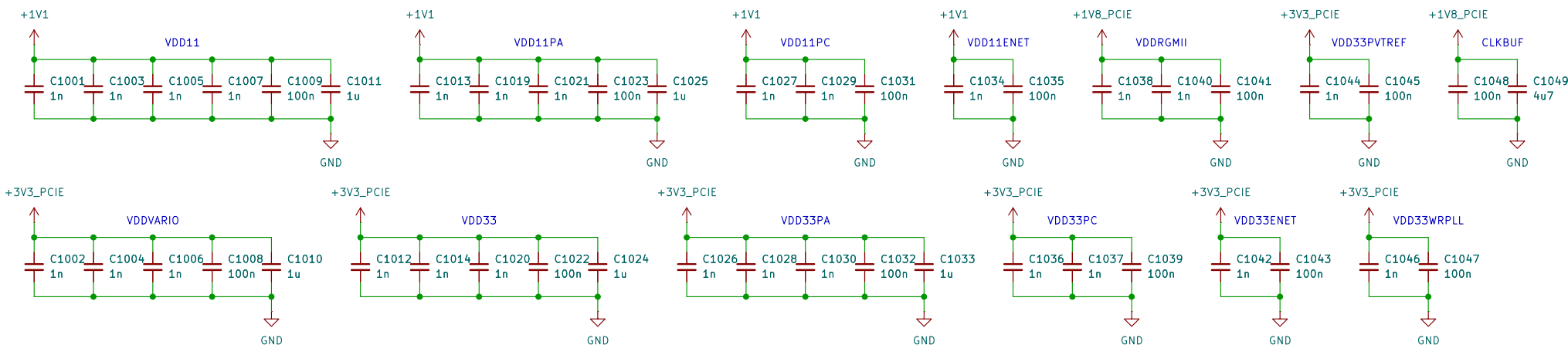
Clock Buffer

The output from the buffer is LP HCSL which is internally terminated meaning no external termination components are required.

PCIe Switch & Ethernet MAC



8KB EEPROM



Sheet: /PCIe Switch/
File: pcie-switch.kicad_sch

Title:

Size: A3

Date:

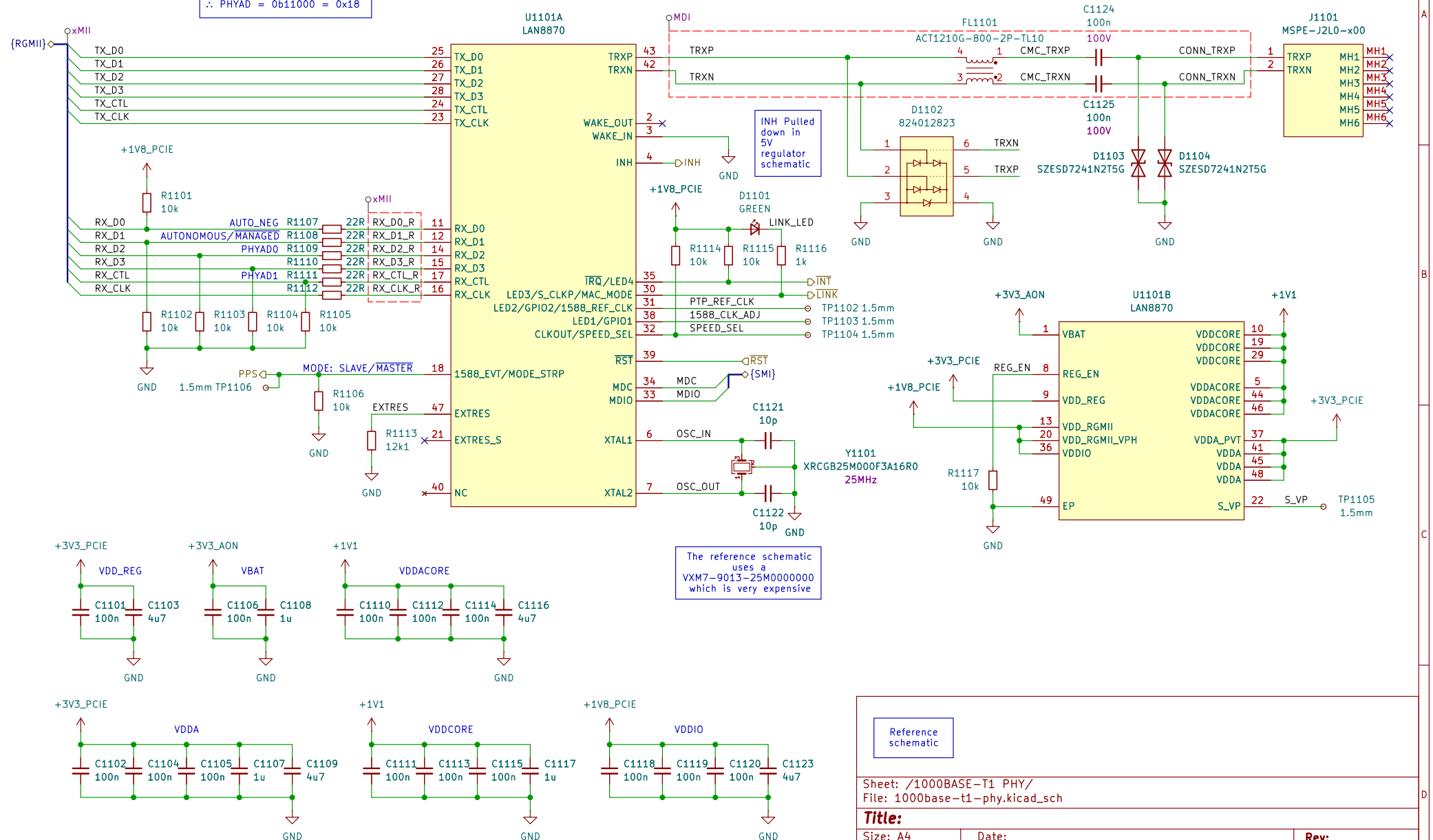
Rev:

KiCad E.D.A. 10.0.0

Id: 10/12

Single Pair Ethernet PHY & Connector

```
When in managed mode:
- SPEED_SEL = PHYAD4
- AUTO_NEG = PHYAD3
- MODE = PHYAD2
∴ PHYAD = 0b11000 = 0x18
```



Reference schematic

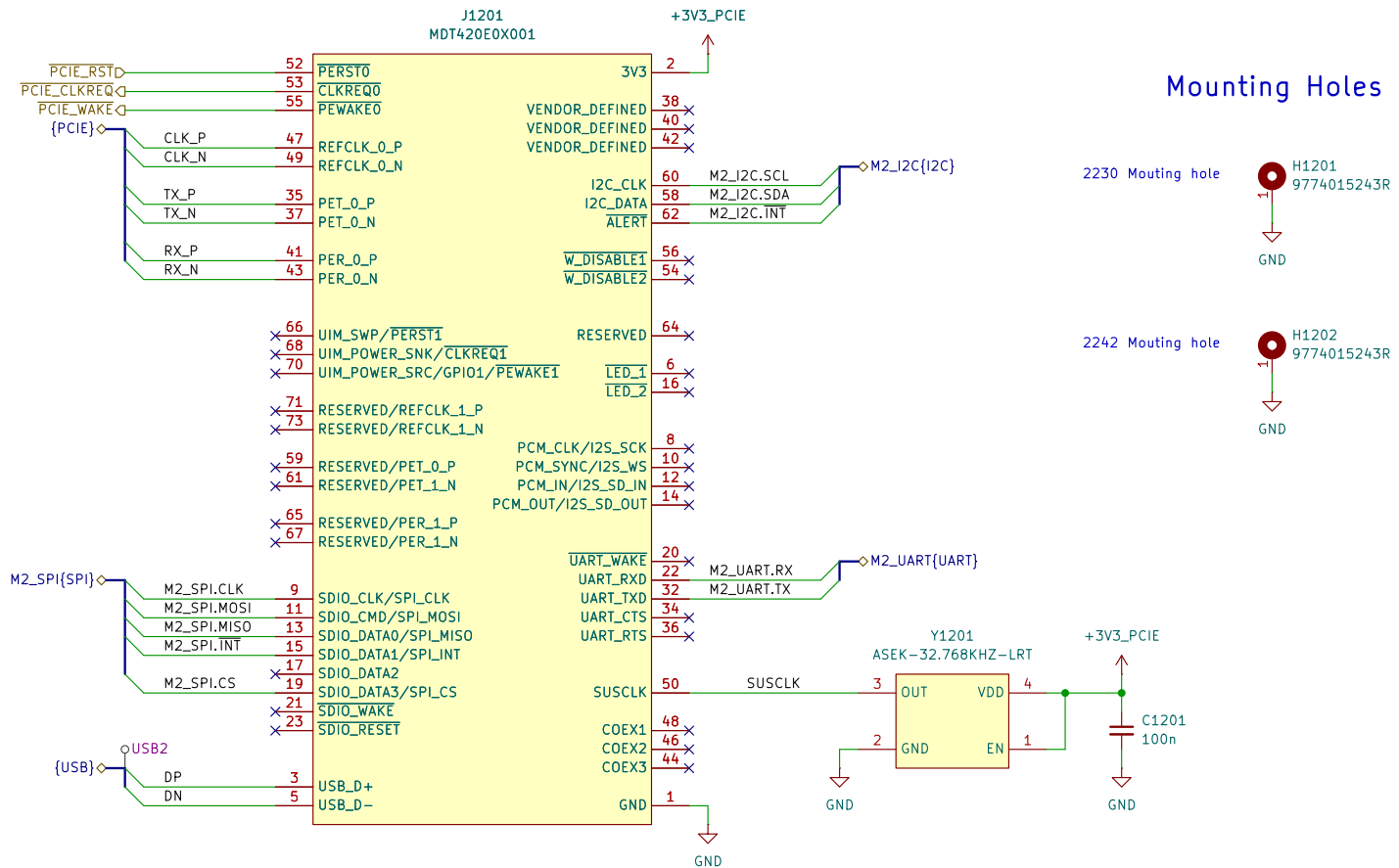
Sheet: /1000BASE-T1 PHY/
File: 1000base-t1-phy.kicad_sch

Title:

Size: A4	Date:
KiCad E.D.A. 10.0.0	

Rev:
Id: 11/12

M.2 E-Key



Sheet: /M.2 E-Key/
File: m-2-e-key.kicad_sch

Title:

Size: A4	Date:
KiCad E.D.A. 10.0.0	

Date:

Rev:
Id: 12/12